College of Computer & Information Sciences
Computer Science
3333 Regis Boulevard
Denver, CO 80221

SYLLABUS

COURSE TITLE: Advanced Programming and Algorithms

COURSE NUMBER: CS 372

COURSE DESCRIPTION:

CS372 – ADVANCED PROGRAMMING AND ALGORITHMS (3). Develops advanced programming skills. Presents advanced features of program design and analysis, group programming, and large projects design and implementation. Introduces analysis of computer algorithms from an efficiency point of view.

PREREQUISITE COURSES:

CS208 Computer Science Fundamentals
and
CS362 Data Structures

In order to successfully participate in this course, students are expected to have completed the course prerequisites. The prerequisites for CS372 are CS208 and CS362, and the prerequisite for CS362 is CS361. Therefore, students ***must have completed CS208, CS361 and CS362*** (or their equivalents) and have a working knowledge of the following topics within each course:

1. Practice with problem definition, solution construction and algorithmic development using top-down design techniques (CS208/*CS361/CS362*).

2. Ability to implement code with C++ control structures used for decisions and iteration, including: if, while, do-while, for, and switch (*CS361*).

3. Understanding of modular code design with C++ functions, and proper parameter passing, using both pass-by-value and pass-by-reference (*CS361/CS362*).

4. Practice with using files to store and access data (*CS362*).

5. Ability to implement code with C++ data structures, including: enumerated types, arrays, structures/records, and linked lists, and the ability to choose appropriate data structures to store data for a particular programming problem (*CS362*).

CS372 is challenging and covers a substantial amount of material at a rapid pace. The only way to succeed in programming is to practice.

As you know by now, programming can require a ***significant*** amount of time each week. There will also be **team projects** in this course, which will demand timeliness.

If you are not willing or able to spend the necessary time for programming and team coordination, please reconsider whether this is the correct time to attend this class.

COURSE OUTCOMES:

Upon completion of this course, learners should be able to:

1. Describe operations on and common applications for linked lists, hash tables, binary search trees, stacks, and queues, and implement them in a high-level programming language.

2. Explain the use of big-O notation to describe the amount of work done by an algorithm and to determine the bounds on time complexity.

3. Compare and contrast various searching methods and their computational efficiencies, and design and implement a hashing function and hash table collision-resolution algorithm appropriate to a specific search application.

4. Describe and implement the most common *quadratic* and *O(N log N)* sorting algorithms, including selection, insertion, bubble, shell, merge and quick sorts, and compare their computational efficiencies and relative advantages.

5. Discuss factors other than computational efficiency that influence the choice of algorithms, and be able to evaluate, select and justify the selection of appropriate algorithms for a common programming problems.

6. Describe the divide-and-conquer approach, and implement, test and debug recursive algorithms to solve programming problems.

7. As a team member, work effectively to design and implement a programming solution, and demonstrate the central elements of team building and team management.

COURSE MATERIALS

Required Texts:

Malik, D.S. (2010). *Data Structures using C++* (2nd edition). Boston, MA: Course Technology; ISBN-13: 978-0-324-78201-1.

Technology Tools (i.e. Required Hardware/Software):

1. A PC-compatible computer system running an MS Windows operating system.

2. A C++ Compiler -- the preferred compiler is the free **Dev-C++** compiler

   The newest version of Dev-C++ (Orwell), can be downloaded from:
   http://sourceforge.net/projects/orwelldevcpp/files/Setup%20Releases/

   Download the most recent version:
   Dev-Cpp x.x.x MinGW x.x.x Setup.exe (32-bit version)
   OR
   Dev-Cpp x.x.x TDM-GCC x64 x.x.x Setup.exe (64-bit version)

   WARNING: If you had an older version loaded on your computer from an earlier class, such as the Bloodshed Dev-C++ compiler, ***you must uninstall it*** before installing the newer version.

   Other ANSI standard C++ compilers can be used, but the facilitator will be most familiar with the Dev-C++ compiler listed above, and will most likely use the Dev-C++ compiler to grade your programs. If a student chooses to use another compiler, it is the student's responsibility to obtain any help needed to use the software, and to insure that all assigned programs compile and run correctly on the Dev-C++ compiler. For team projects, the Dev C++ compiler will be the default, unless **all** team members agree on using another compiler.

   As with most of Regis learning activities, using various software applications to accomplish assignments requires students to exercise a great deal of responsibility for learning how to successfully operate the software applications. This class assumes a working knowledge of the C++ compiler from CS362. There are, however, many quality books on the market that support novice users if a student needs them.

PRE-ASSIGNMENT:

Students will read the following in the required textbook.

Students will be PREPARED to ASK QUESTIONS on unclear areas and TO RESPOND to questions about information in the assigned reading.

- Chap 1: pgs 1 – middle of 8 (SW development)
- Chap 3: pgs 132 – top of 155 (pointers)
- Chap 5: pgs 266 – 320 (linked lists)
- Chap 7, all (stacks)
- Chap 8, all (queues)
- ***Optional:*** Chap 1: pgs 18-51 (classes)

COURSE ASSIGNMENTS AND ACTIVITIES

Week	Learning Topics	Readings in required textbook*	Graded Assignments and Assessments**
1	LT 1: C++ Concepts LT 2: ADTs and Linked Lists LT 11: Stacks and Queues	Chap 1: pgs 1 – middle of 8 (SW development); Chap 3: pgs 132 – top of 155 (pointer review); Chap 5: pgs 266 – 310 (linked list review); Chap 7, all (stacks); Chap 8, all (queues) ***Optional:*** Chap 1: 18-51 (classes)	Forum Participation Programming Assn 1 (individual – stacks and queues)
2	LT 2 (continued): ADTs and Linked Lists (variations) LT 3: Algorithm Efficiency and Big-O Notation	Chap 5: pgs 310 – 320 (doubly linked lists) Chap 1: pgs 8 – 17 (big-O)	Forum Participation Programming Assn 2 (individual - doubly linked lists)
3	LT 4: Large Projects LT 5: Basic Searching Algorithms LT 6: Searches with Hashing	Chap 9: all (searching/hashing)	Forum Participation Programming Assn 3 Design (team – searching/hashing)
4	LT 6 (continued): Searches with Hashing (analysis) LT 7: Recursive Algorithms	Chap 6: all (recursion)	Forum Participation Programming Assn 3 (team – searching /hashing) Midterm Exam
5	LT 9: Simple Sorting Algorithms LT 1: C++ Concepts (ptrs to functions)	Chap 10: pgs 534 – top of 552 (comparison sorts) Appendix F: pgs 825 – 826 (insertion sort analysis)	Forum Participation Programming Assn 4 Design (team – sorting)
6	LT 10: Complex Sorting Algorithms	Chap 10: 552 – 567 (complex sorts) Appendix F: pgs 826 – 832 (quick sort analysis)	Forum Participation Programming Assn 4 (team – sorting)
7	LT 8: Binary Search Trees	Chap 11: pgs 600 - 608 (binary trees), 616 – top of 628 (binary search trees)	Forum Participation
8	LT 12: Advanced Algorithm Issues		Forum Participation Programming Assn 5 (individual – binary trees) Final Exam – 20%

****Alternate Reading Assignments:*** Alternate reading assignments for the previously required textbooks will be available in your facilitator's syllabus, in case you purchased them previously.

*****Note to Classroom sections only:*** Exact due dates for programming assignments may be modified from what is indicated in the **Course Overview Grid**. Your facilitator's syllabus, handed out the first night of class, will indicate any changes.

Summary of Assignments and Percentage Weight towards course grade

Assignment	Value (percent of overall course grade)
Programming Assignments (5)	
Program 1 (individual)	10 %
Program 2 (team)	10 %
Program 3 (individual)	10%
Program 4 (team)	10%
Program 5 (individual)	10 %
Programming Total	**50 %**
Midterm Exam	**20 %**
Final Exam	**20 %**
Classroom: Class Participation/Pop Quizzes ***Online:*** Forum Participation	**10 %**
Totals	100 %

Programming Assignments

Each programming assignment will involve designing and writing a program, using the concepts discussed in the book and class, and may also include submission of analyses, teammate evaluations, structure charts, test data and/or test plans. **Note:** Programs that do not compile, are not modular, or that use any global variables, will NOT be accepted.

One of the goals of this class is to learn how to design and implement a program as part of a programming team. Therefore, some of your assignments will be team-programming assignments. Note that although students will work in teams, the grades will be individual.

CS372 Programming Assignment coverage and Late policies

Program #1 will be an *individual* assignment that covers stacks and queues.
(will not be accepted more than one day late)

Program #2 will be an *individual* assignment that covers doubly linked lists.
(will not be accepted more than one day late)

Program #3 will be a *team* assignment that covers data storage using hashing, and comparison of searches using various hashing collision resolution methods.
(will not be accepted more than one day late)

Program #4 will be a *team* assignment on timing and comparison of sorting methods.
(will not be accepted more than 4 days late)

Program #5 will be an *individual* assignment that covers binary search trees.
(will not be accepted more than 4 days late)

Late Policy for CS372 Programming Assignments

Due to the requirement to forms teams in this class, there will be a strict adherence to the late policy. Programming assignments will be graded, and then 2% will be deducted *for each day* the assignment is late, up to the maximum allowed.

Programming assignments turned in more than 1 day late (for assns 1-3) or 4 days late (for assns 4-5) will be given a grade of **zero**, and no feedback will be given.

Exams

There will be a midterm exam and a final exam. Exam questions will be cumulative, taken from reading assignments and course content. ***Exams will not be accepted late.***

Participation

Class participation/effort is important because we can all learn from each other. Your participation points can make a difference in the final grade. Participation means:

1. Present in class every session (classroom)/in the forum every week (online)
2. a. Effectively responds to questions from the facilitator (classroom)
   b. Regularly checks forum and posts all required items by the deadlines (online)
3. Interacts/replies to other students in classroom/forum discussions.

Late Assignment Policy for CS372 Online Discussions

Late discussion posts will have 10% deducted for each day the post is late, up to one week late. Any discussion assignment posted more than one week late will be given a grade of **zero**.

Academic Integrity

All students will be subject to Regis University's Academic Integrity policies.

Plagiarism includes submitting code obtained from any person or from any internet web source. All work submitted in CS372 must be your own.

In cases of suspected cheating or plagiarism,the instructor will discuss the matter with the student(s) involved. The instructor reserves the right to question any student orally or in writing about any assignment, and to use the evaluation of the student's understanding of the assignment and of the submitted solution as evidence of cheating.

Final Course Grades

CC&IS Grading Scale

Letter Grade	*Percentage*	*Grade Point*
A	93 to 100	4.00
A–	90 to less than 93	3.67
B+	87 to less than 90	3.33
B	83 to less than 87	3.00
B–	80 to less than 83	2.67
C+	77 to less than 80	2.33
C	73 to less than 77	2.00
C–	70 to less than 73	1.67
D+	67 to less than 70	1.33
D	63 to less than 67	1.00
D–	60 to less than 63	.67
F	Less than 60	0